

Proj1：数据的捕获与构建

1 Step2：数据获取与初始入库

1.1 数据获取

数据来源：Kaggle 开源数据集 Brazilian E-Commerce Public Dataset by Olist。原始数据以 8 张 CSV 表格形式提供，本项目使用其中 5 张：

文件名	说明	主要字段
olist_customers_dataset.csv	客户表	customer_id, customer_unique_id
olist_orders_dataset.csv	订单表	order_id, customer_id, 购买/送达/预计送达时间戳
olist_order_items_dataset.csv	订单明细表	order_id, order_item_id, product_id, price, freight_value
olist_products_dataset.csv	商品表	product_id, product_category_name
olist_order_reviews_dataset.csv	评价表	review_id, order_id, review_score

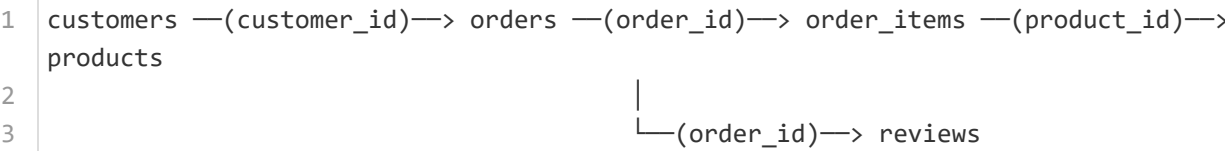
数据规模：原始订单记录约 99,441 条，订单明细约 112,000 条，商品约 32,000 种，客户约 96,000 位。

1.2 数据清洗

清洗工作由 Python 脚本 `clean_flat_table.py` 完成，使用 `pandas` 库进行数据处理。脚本的核心流程如下：

1. 数据合并流程

脚本从 `archive/` 目录读取上述 5 个 CSV 文件，仅选取分析所需的列，然后通过 `inner join` 逐步合并：



合并后得到一张包含所有维度的大宽表，每行代表一个订单明细与评价的组合。

2. 缺失值处理策略

字段	处理方式	理由
order_delivered_customer_date	丢弃该行（删除未送达订单）	三个分析目标均依赖已送达订单的时间数据，未送达订单无法参与物流时效分析
product_category_name	填充为 'Unknown'	保留交易记录完整性，避免因类别缺失而丢失价格、运费等有效数据
review_score	填充为 3（中位值）	评分 1-5 分，3 为中间值，减少对均值的极端拉偏

3. 脏数据处理

三个时间戳字段（`order_purchase_timestamp`、`order_delivered_customer_date`、`order_estimated_delivery_date`）通过 `pd.to_datetime(errors='coerce')` 统一解析，输出时格式化为 `%Y-%m-%d %H:%M:%S`，确保时间格式一致。

4. 清洗结果统计

指标	数值
原始订单数	99,441
丢弃未送达订单数	2,965
清洗后订单数	96,476
商品类别原始缺失数	610
最终宽表行数	110,012
最终宽表列数	12

最终宽表 `cleaned_flat_table.csv` 包含以下 13 列: `id`、`order_id`、`order_item_id`、`customer_unique_id`、`order_purchase_timestamp`、`order_delivered_customer_date`、`order_estimated_delivery_date`、`product_id`、`product_category_name`、`price`、`freight_value`、`review_id`、`review_score`。

行数 (110,012) 大于订单明细数, 原因是部分订单存在多条评价记录, 合并后保留了每条评价与对应订单明细的组合。每行通过 `(order_id, order_item_id, review_id)` 唯一标识。

5. 运行方式

```
1 | python clean_flat_table.py --input-dir archive --output cleaned_flat_table.csv
```

1.3 初始数据库实现

1.3.1 openGauss 部署

采用 Docker Desktop 部署 openGauss 官方镜像。容器启动命令如下:

```
1 | docker run \
2 |   --env=GS_PASSWORD=MyGauss@123 \
3 |   --volume=E:\Docker\OpenGauss:/var/lib/opengauss \
4 |   -p 15432:5432 \
5 |   -d opengauss/opengauss:latest
```

1.3.2 初始表结构设计

本阶段采用非规范化的大宽表方案, 将清洗后的所有数据存入单张表中, 对应 DDL 脚本为

```
schema.sql:
1 | CREATE TABLE cleaned_flat_table (
2 |     id SERIAL PRIMARY KEY,
3 |     order_id VARCHAR(32) NOT NULL,
4 |     order_item_id INTEGER NOT NULL CHECK (order_item_id >
5 | 0),
6 |     customer_unique_id VARCHAR(32) NOT NULL,
7 |     order_purchase_timestamp TIMESTAMP NOT NULL,
8 |     order_delivered_customer_date TIMESTAMP NOT NULL,
9 |     order_estimated_delivery_date TIMESTAMP NOT NULL,
10 |     product_id VARCHAR(32) NOT NULL,
11 |     product_category_name VARCHAR(100) NOT NULL DEFAULT 'Unknown',
12 |     price NUMERIC(10,2) NOT NULL CHECK (price >= 0),
13 |     freight_value NUMERIC(10,2) NOT NULL CHECK (freight_value >=
14 | 0),
15 |     review_id VARCHAR(32) NOT NULL,
```

```

14         review_score          INTEGER          NOT NULL DEFAULT 3
15                                CHECK (review_score BETWEEN 1 AND 5),
16         UNIQUE (order_id, order_item_id, review_id)
17 );

```

完整性约束设计说明：

- **主键：** `id SERIAL`，自增整数代理主键，索引高效，适合过渡性宽表。
- **UNIQUE：** `(order_id, order_item_id, review_id)` 组合唯一约束，保证记录无重复。一个订单可包含多个商品项，一个订单可有多条评价，三者联合唯一标识每条记录。
- **NOT NULL：** 所有列均设为 NOT NULL，因清洗脚本已处理所有缺失值。
- **DEFAULT：** `product_category_name` 默认 'Unknown'，`review_score` 默认 3，与清洗逻辑保持一致。
- **CHECK：** `price >= 0` 和 `freight_value >= 0`（金额不为负）；`order_item_id > 0`（项号从 1 开始）；`review_score BETWEEN 1 AND 5`（评分范围限制）。

1.3.3 导入 Navicat

id # int4	order_id # int4	order_item_id # int4	customer_unique_id # varchar(32)	order_purchase_time # timestamp(6)	order_delivered_cust # timestamp(6)	order_estimated_deliv # timestamp(6)	product_id # varchar(32)	product_category_na # varchar(100)	price # numeric	review_id # varchar(32)	freight_value # numeric(10,2)	review_score # int4
1	e481f51cbd	1	7c396fd4830fd04220f75	2017-10-02 10:56:33	2017-10-10 21:25:13	2017-10-18 00:00:00	87285b348845	utilidades_domesticas	29.99	a54f0611adc9ed256b57	8.72	4
2	53cdb2fcb8	1	a0f7308b275d755c9edb	2018-07-24 20:41:37	2018-08-07 15:27:45	2018-08-13 00:00:00	595fac2a385ac	perfumaria	118.70	8d5266042046a06655c	22.76	4
3	47770eb91c	1	3a653a41f6f9f3d2a113	2018-08-08 08:38:49	2018-08-17 18:06:29	2018-09-04 00:00:00	aa4383b373c6	automotivo	159.90	e73b67b67587f7644d5t	19.22	5
4	949d5b44d0	1	71c42cf63193a1473d2e	2017-11-18 19:28:06	2017-12-02 00:28:42	2017-12-15 00:00:00	d0b61bfb1de8	pet_shop	45.00	359d03e676b3c069f62c	27.20	5
5	ad21c59c08	1	72632f0f9dd3fdee390c	2018-02-13 21:18:39	2018-02-16 18:17:02	2018-02-26 00:00:00	65266b2da20d	papelaria	19.90	e50934924e227544ba8	8.72	5
6	a4591c265e	1	80bb27c7c16e8f973207	2017-07-09 21:57:05	2017-07-26 10:57:55	2017-08-01 00:00:00	060cb19345d9	automotivo	147.90	89b738e70a1ce346db2t	27.36	4
7	6514b8ad8c	1	932afa1e708222e5821d	2017-05-16 13:10:30	2017-05-26 12:55:51	2017-06-07 00:00:00	4520766ec412	automotivo	59.99	07d67dd06ed5f88bef1t	15.17	5
8	76c6e86628	1	39382392765bdc7481c	2017-01-23 18:29:09	2017-02-02 14:08:10	2017-03-06 00:00:00	ac1789e492dc	moveis_decoracao	19.90	fc4af8aaebc3f1a3cd18t	16.05	1
9	e69bf5eb8	1	299905e3934e9e181bft	2017-07-29 11:55:02	2017-08-16 17:14:30	2017-08-23 00:00:00	9a78fb9862b1c	moveis_escritorio	149.99	abc5655186d40772bd6t	19.77	5
10	e6ce16cb79	1	f2a85dec752b8517b5e5	2017-05-16 19:41:10	2017-05-29 11:18:31	2017-06-07 00:00:00	08574b074924	ferramentas_jardim	99.00	15898b543726a832d41t	30.53	1
11	e6ce16cb79	2	f2a85dec752b8517b5e5	2017-05-16 19:41:10	2017-05-29 11:18:31	2017-06-07 00:00:00	08574b074924	ferramentas_jardim	99.00	15898b543726a832d41t	30.53	1
12	34513ce0c4	1	782987b81c92239d922t	2017-07-13 19:58:11	2017-07-19 14:04:48	2017-08-08 00:00:00	f7e0fa61b5386	informatica_acessorios	98.00	a932e1fddc0c17fc26de	16.13	4
13	82566a660a	1	e97109680b052ee88d	2018-06-07 10:06:19	2018-06-19 12:05:52	2018-07-18 00:00:00	72a97c271b2e	perfumaria	31.90	ed220e853e96b629118t	18.23	5
14	5ff96c15d0t	1	e2dfa3127fedbbca9707t	2018-07-25 17:44:10	2018-07-30 15:52:25	2018-08-08 00:00:00	10adb53d8faa	cama_mesa_banho	19.90	346e42116863ae64d51c	12.80	5
15	432aa2f21d8t	1	04cf8185c71090d28baa	2018-03-01 14:14:28	2018-03-12 23:36:26	2018-03-21 00:00:00	72d3bf1d3a79t	brinquedos	38.25	8f39f09c22b1cebbe3eb	16.11	4
16	dcb36b511f	1	ccaf1c3f270410521c3c	2018-06-07 19:03:12	2018-06-21 15:34:32	2018-07-04 00:00:00	009c09f43998t	perfumaria	132.40	5a86084374b52bec5d7t	14.05	5
17	116f060934	1	6087cf70f4833cf2db63	2017-12-26 23:41:31	2018-01-08 22:36:36	2018-01-29 00:00:00	a47295965bd0	telefonias	27.99	0c00083494717bc3697t	15.10	5
18	85ce859fd6	1	d0ffa17468fcc46b8f6c5t	2017-11-21 00:03:41	2017-11-27 18:28:00	2017-12-11 00:00:00	cce679660cb6	cama_mesa_banho	17.90	70c498f8918af37d644c	11.85	5
19	83018ec114	1	634f09f6075fe9032e6c1	2017-10-26 15:54:26	2017-11-08 22:22:00	2017-11-23 00:00:00	c35498fbb435t	telefonias	76.00	219cf59cf889bc85babbt	16.97	4
20	203096f03d	1	d699688533772c15a061	2017-09-18 14:31:30	2017-10-09 22:23:46	2017-09-28 00:00:00	5ac9d9e379c6t	beleza_saude	109.90	38cae21b1b57a959594t	8.96	2
21	f848643eec	1	68954feaafe4dd638f3bc	2018-03-15 08:52:40	2018-03-19 18:08:32	2018-03-29 00:00:00	2b4609f8948bt	beleza_saude	79.99	f3d7d8d2a38f81f086bcb	8.91	5
22	2807d0e504	1	b8df986511d928829c31	2018-02-03 20:37:35	2018-02-08 16:13:46	2018-02-21 00:00:00	6893767814d1t	eletronicos	9.50	e905f4e4e372b175115c	7.78	3
23	95266dbf7	1	451e48381edab7f1f6db	2018-01-08 07:55:29	2018-01-26 17:32:38	2018-02-21 00:00:00	bb7181410b4e	bebes	129.99	1d507207fdd8c75b7853t	57.58	4
24	f3e7c35915t	1	9c9242ad7f1b5d2926ea	2018-08-09 11:44:40	2018-08-13 18:24:27	2018-08-17 00:00:00	e99d69fefe684t	papelaria	89.90	9a55cb5c8f8ad47581d	14.21	5
25	fbf9ac61453	1	a25d5f94840d3c6a1a49	2018-02-20 23:46:53	2018-03-21 22:03:54	2018-03-12 00:00:00	7b717060aa78	cool_stuff	109.90	6a1a8e54de03ab98e6et	15.53	2
26	acce194856	1	576ea0cab426cd8a00f	2018-06-04 00:00:13	2018-06-16 15:20:55	2018-07-18 00:00:00	d70f38e77f9c6t	bebes	90.90	2d4303fd561eb4ab51c8	48.64	1
27	acce194856	2	576ea0cab426cd8a00f	2018-06-04 00:00:13	2018-06-16 15:20:55	2018-07-18 00:00:00	9451e630d725t	bebes	39.50	2d4303fd561eb4ab51c8	48.64	1
28	dd78f560c2	1	f4a7ef6bd931f83d75d8t	2018-03-12 01:50:26	2018-03-21 14:41:50	2018-03-28 00:00:00	00baba5b58e2	perfumaria	47.90	679046d7df350a6376et	12.79	5
29	91b2a010e1	1	bd136608b9c6033892cet	2018-05-02 11:45:38	2018-05-16 20:56:24	2018-05-23 00:00:00	ba74c6b75d2a	relorios_presentes	178.99	8a4283cd714fca8f3da9	13.69	4
30	ecab90c993	1	1428917cd397d4f9ac0f	2018-02-25 13:50:30	2018-03-27 23:29:14	2018-04-13 00:00:00	c0db539123a4t	ferramentas_jardim	30.10	0153550f6dd1a6bdfc33t	33.24	4
31	f70a0aff17d	1	1974875b4a1d2e2ee6df	2017-08-10 11:58:33	2017-08-18 14:28:02	2017-08-23 00:00:00	cafd558df4c3c	bebes	279.00	2f111691fcd93d88f3aae	34.19	2
32	1790eea0b5	1	a6fecd9f434474c6fcd8	2018-04-16 21:15:39	2018-05-05 12:28:34	2018-05-15 00:00:00	2d8f2be4f087et	climatizacao	186.90	764c587207b9589949ak	38.00	1
33	989225ba6c	1	37363700139c1ae87f3b	2017-12-12 13:56:04	2018-01-03 18:03:36	2018-01-08 00:00:00	6b64362e8989	moveis_decoracao	49.00	3ee0f16b91a29a331bb	14.10	4
34	d887b52c65	1	a2649503b92028291f01	2018-02-03 12:38:58	2018-02-22 00:07:55	2018-03-07 00:00:00	39a9942865c0t	brinquedos	84.90	57bec0c29a5113d8344	15.35	5
35	b276e4f8c0	1	6cbe8a392b76916e84c2	2018-07-29 23:34:51	2018-07-31 22:48:50	2018-08-06 00:00:00	c6c1f263e076t	perfumaria	179.00	e71c2035f3e772a0e233t	9.41	5

1.4 基础查询验证

连接 Navicat，执行以下 SQL 语句验证数据正确入库：

查询 1：验证总行数

```
1 SELECT COUNT(*) AS total_rows FROM cleaned_flat_table;
```

保存 查询创建工具 美化 SQL 代码段 询问 AI

My_openGauss ecommerce_db public 运行 停止 解释

1 SELECT COUNT(*) AS total_rows FROM cleaned_flat_table;

消息 摘要 结果 1

数据 信息

total_rows
110012

查询 2: 验证主键唯一性

```
1 SELECT order_id, order_item_id, review_id, COUNT(*)
2 FROM cleaned_flat_table
3 GROUP BY order_id, order_item_id, review_id
4 HAVING COUNT(*) > 1;
```

1 SELECT order_id, order_item_id, review_id, COUNT(*)
2 FROM cleaned_flat_table
3 GROUP BY order_id, order_item_id, review_id
4 HAVING COUNT(*) > 1;

消息 摘要 结果 1

数据 信息

order_id	order_item_id	review_id	count
(N/A)	(N/A)	(N/A)	(N/A)

查询 3: 验证约束字段范围

```
1 SELECT
2     MIN(price) AS min_price,
3     MAX(price) AS max_price,
4     MIN(freight_value) AS min_freight,
5     MAX(freight_value) AS max_freight,
6     MIN(review_score) AS min_score,
7     MAX(review_score) AS max_score
8 FROM cleaned_flat_table;
```

1	SELECT
2	MIN(price) AS min_price,
3	MAX(price) AS max_price,
4	MIN(freight_value) AS min_freight,
5	MAX(freight_value) AS max_freight,
6	MIN(review_score) AS min_score,
7	MAX(review_score) AS max_score
8	FROM cleaned_flat_table;

消息	摘要	结果 1
----	----	------

数据	信息
----	----

min_price	max_price	min_freight	max_freight	min_score	max_score
.85	6735.00	0.00	409.68	1	5

查询 4：抽样查看数据

```
1 SELECT
2     order_id,
3     order_item_id,
4     customer_unique_id,
5     product_category_name,
6     price,
7     freight_value,
8     review_score,
9     order_purchase_timestamp
10 FROM cleaned_flat_table
11 ORDER BY order_purchase_timestamp DESC
12 LIMIT 10;
```

1	SELECT
2	id,
3	order_id,
4	order_item_id,
5	customer_unique_id,
6	product_category_name,
7	price,
8	freight_value,
9	review_score,
10	order_purchase_timestamp
11	FROM cleaned_flat_table
12	ORDER BY order_purchase_timestamp DESC
13	LIMIT 10;

消息	摘要	结果 1
----	----	------

数据	信息
----	----

id	order_id	order_item_id	customer_unique_id	product_category_name	price	freight_value	review_score	order_purchase_time:
97357	35a972d7f8	1	24ac2b4327e25baf39f21	informatica_acessorios	84.99	8.76	5	2018-08-29 15:00:37
77033	03ef5dedbe	1	b701bebbdf478f550034	artigos_de_festas	24.90	8.33	3	2018-08-29 14:52:00
34884	168626408c	1	7febafa06d9d8f232a900	beleza_saude	45.90	15.39	5	2018-08-29 14:18:28
33296	0b223d92c2	1	5c58de6fb80e93396e2f	moveis_cozinha_area_de	209.00	46.48	5	2018-08-29 14:18:23
33297	0b223d92c2	2	5c58de6fb80e93396e2f	moveis_cozinha_area_de	209.00	46.48	5	2018-08-29 14:18:23
109937	5201848470	1	7a22d14aa3c359923850	brinquedos	63.90	9.20	1	2018-08-29 12:25:59
75005	d03ca98f59	1	0421e7a23f21e5d54efec	alimentos	109.90	9.52	5	2018-08-29 11:06:11
6333	d70442bc5e	1	21dbe8eabd00b34492a	esporte_lazer	6.90	7.39	5	2018-08-29 10:22:35
97707	912859fef5a	1	0c6d7218d5f3fa14514fd	relorios_presentes	169.80	8.45	5	2018-08-29 09:48:09
82626	fb39321145	1	b4dcade04bc548b7e3b1	relorios_presentes	99.00	7.95	1	2018-08-29 09:14:11